

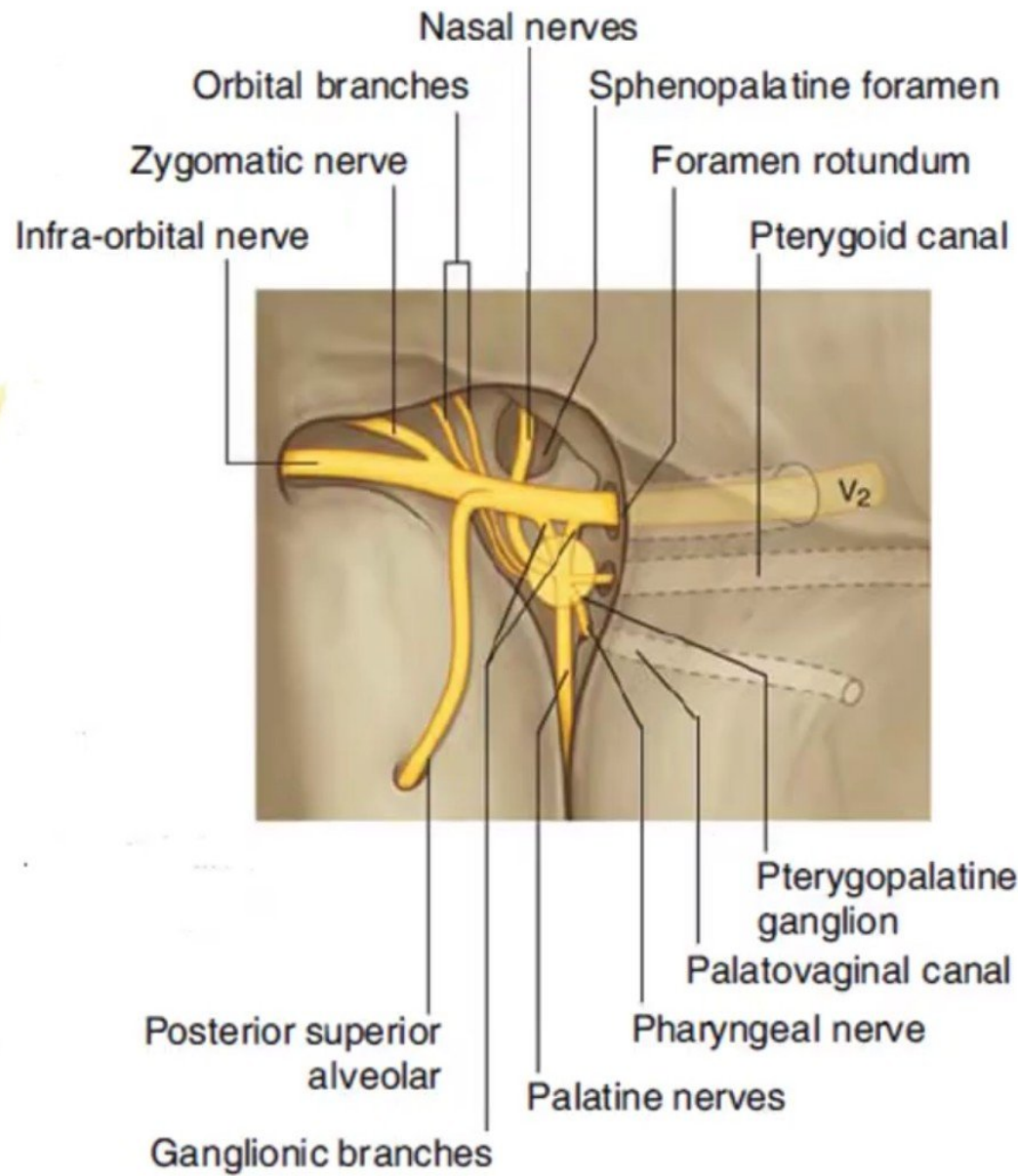
PTERYGOPALATINE FOSSAE

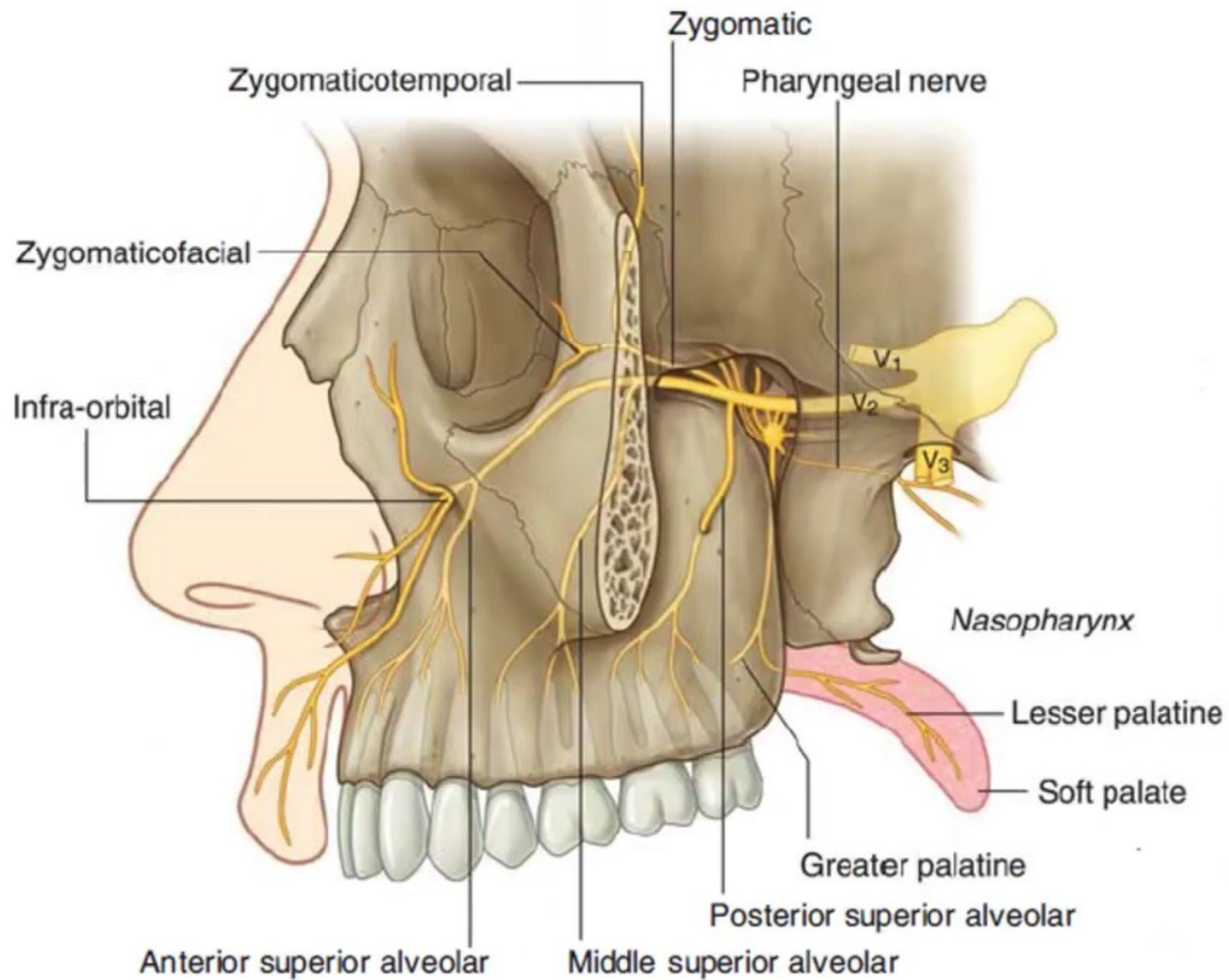
Dr. Osman Ahmed Osman Fadl
Department of Anatomy

Position

- ▶ A small pyramidal space inferior to the orbit.

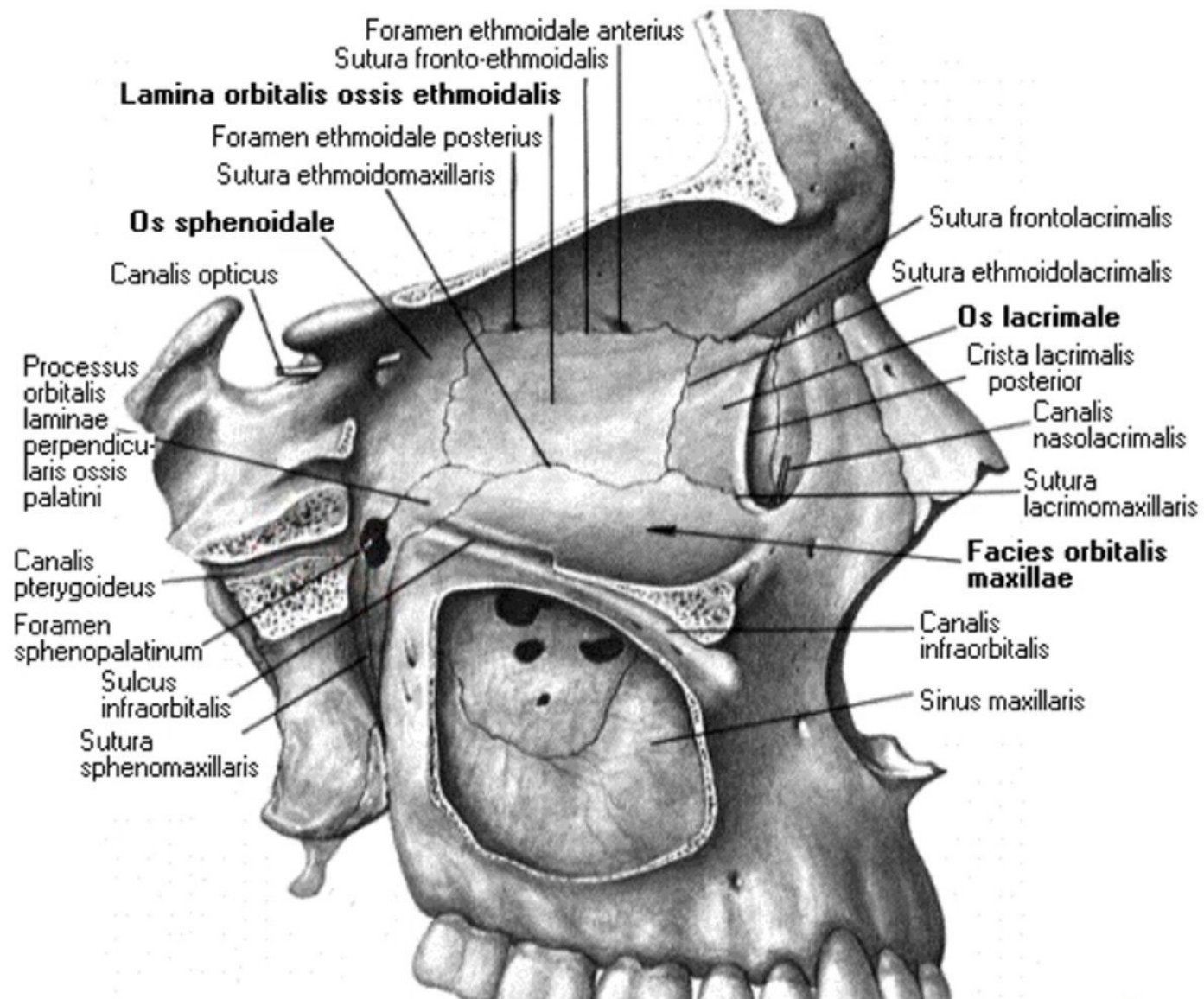






Boundaries

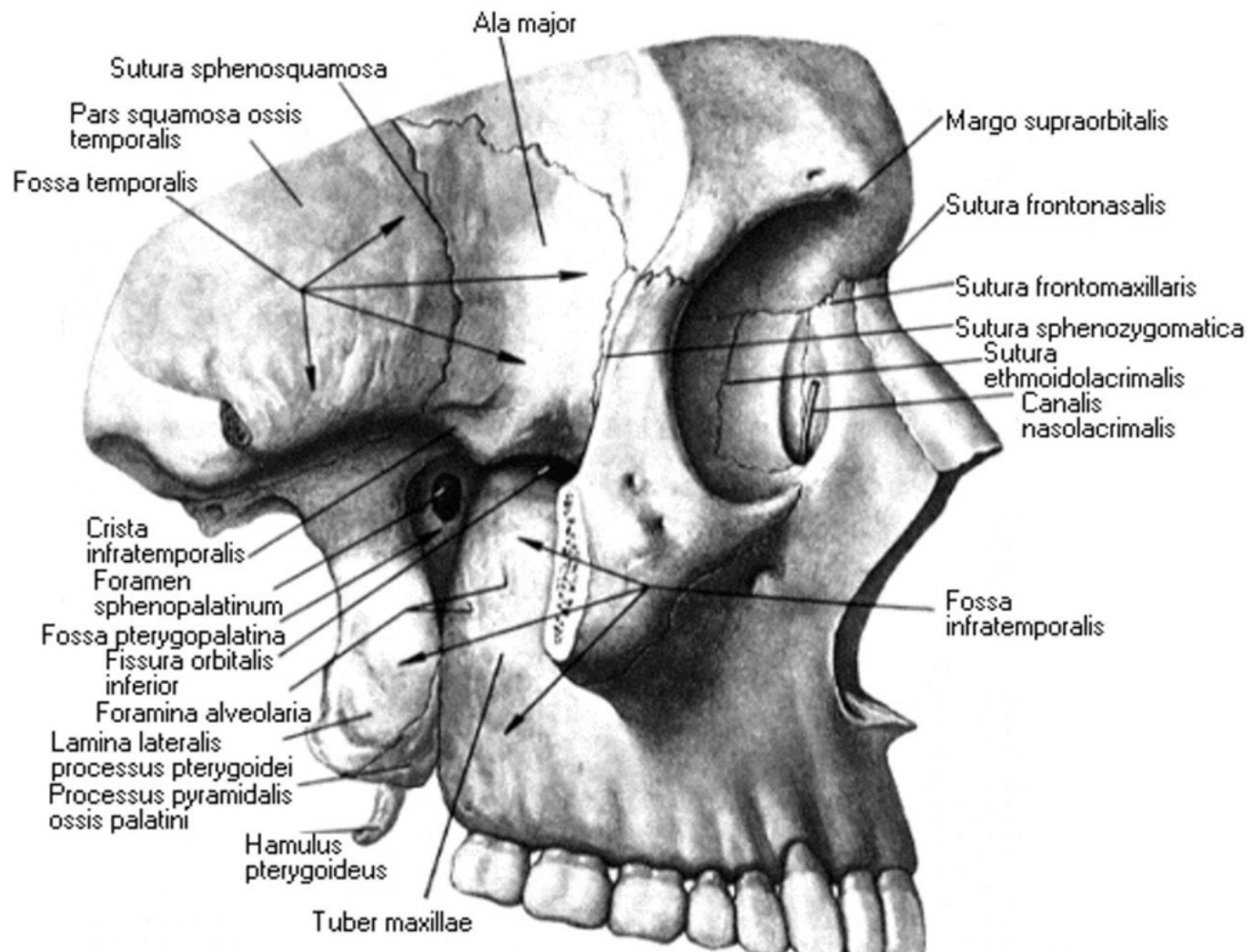
- Anteriorly: posterior aspect of the maxilla.
- Posteriorly: pterygoid process of the sphenoid bone.
- Medially: vertical plate of the palatine bone.
- Laterally: open into pterygomaxillary fissure.

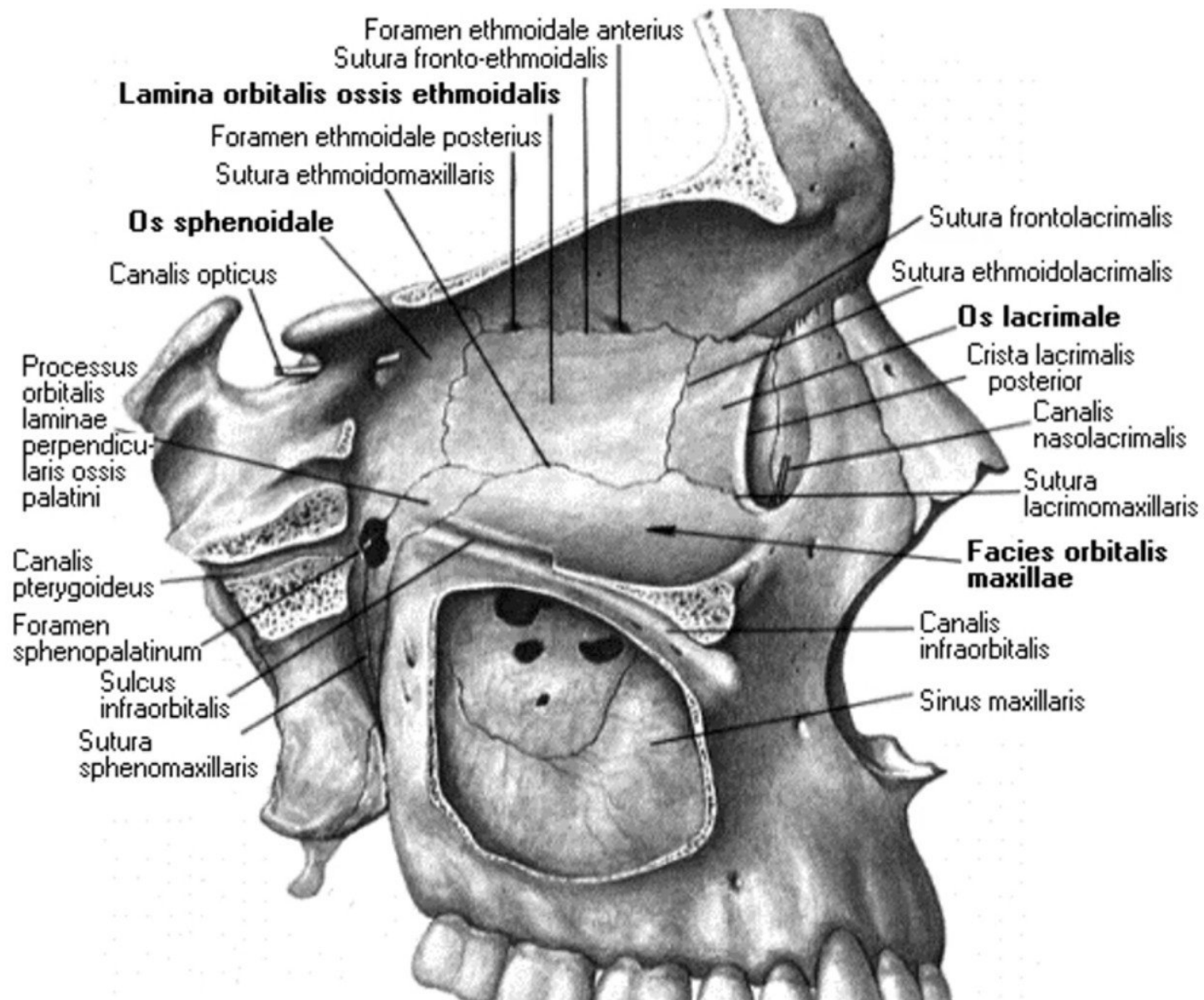




Communications

- ▶ Laterally with the infratemporal fossa through the pterygomaxillary fissure.
- ▶ Medially with nasal cavity through the sphenopalatine foramen.
- ▶ Anterosuperiorly with the orbit through the inferior orbital fissure.
- ▶ Posterosuperiorly with middle cranial fossa through foramen rotundum.





Contents

- ▶ Third part of maxillary artery.
- ▶ Maxillary nerve.
- ▶ Pterygopalatine ganglion.

Pterygopalatine ganglion

▶ Parasympathetic root.

- ▶ From the superior salivary nucleus by the nervus intermedius part of **the facial nerve**, the greater petrosal nerve and the nerve of the pterygoid canal.

▶ Sympathetic root.

- ▶ From **the superior cervical ganglion** by the internal carotid plexus, the deep petrosal nerve and the nerve of the pterygoid canal.

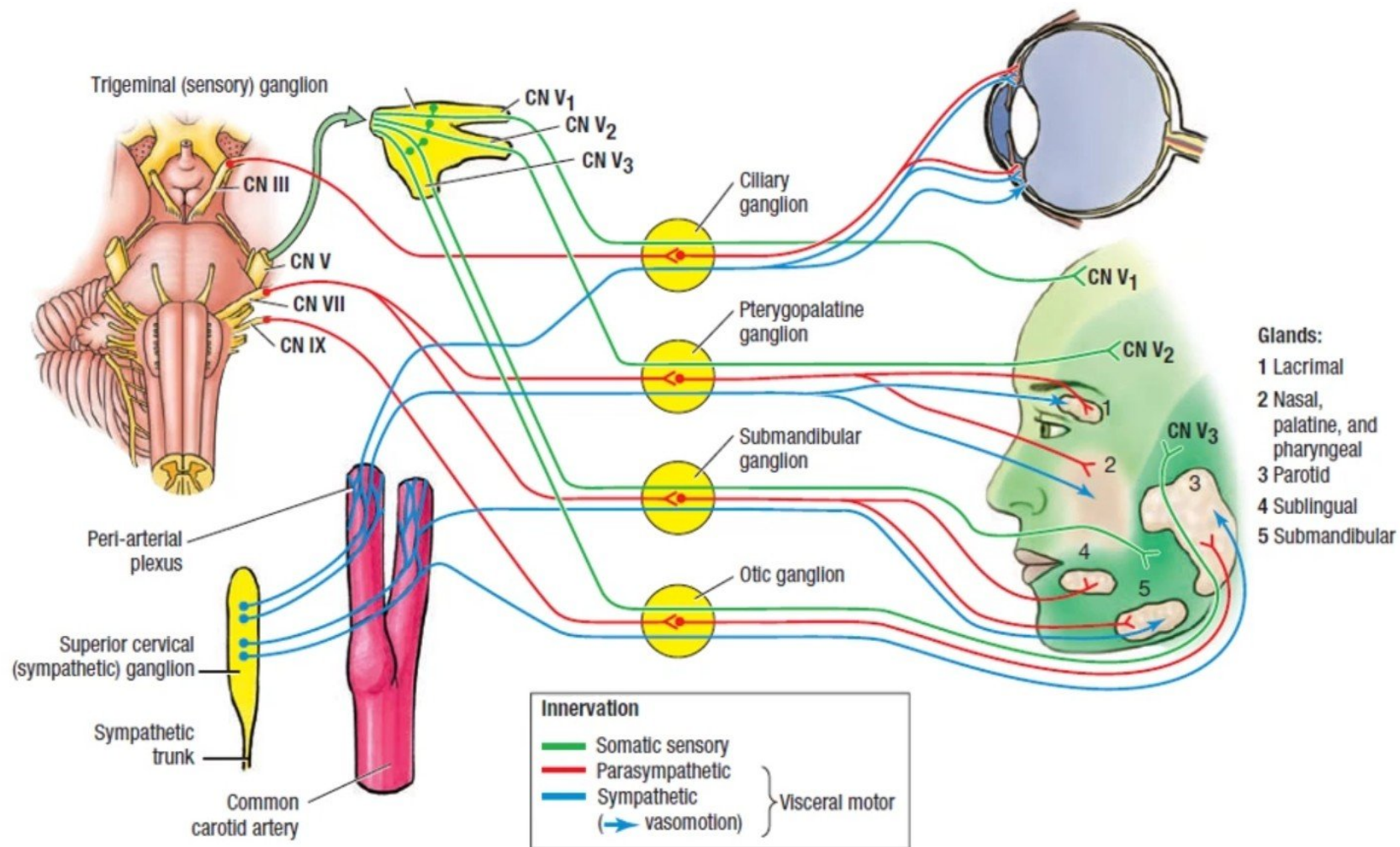
▶ Sensory root.

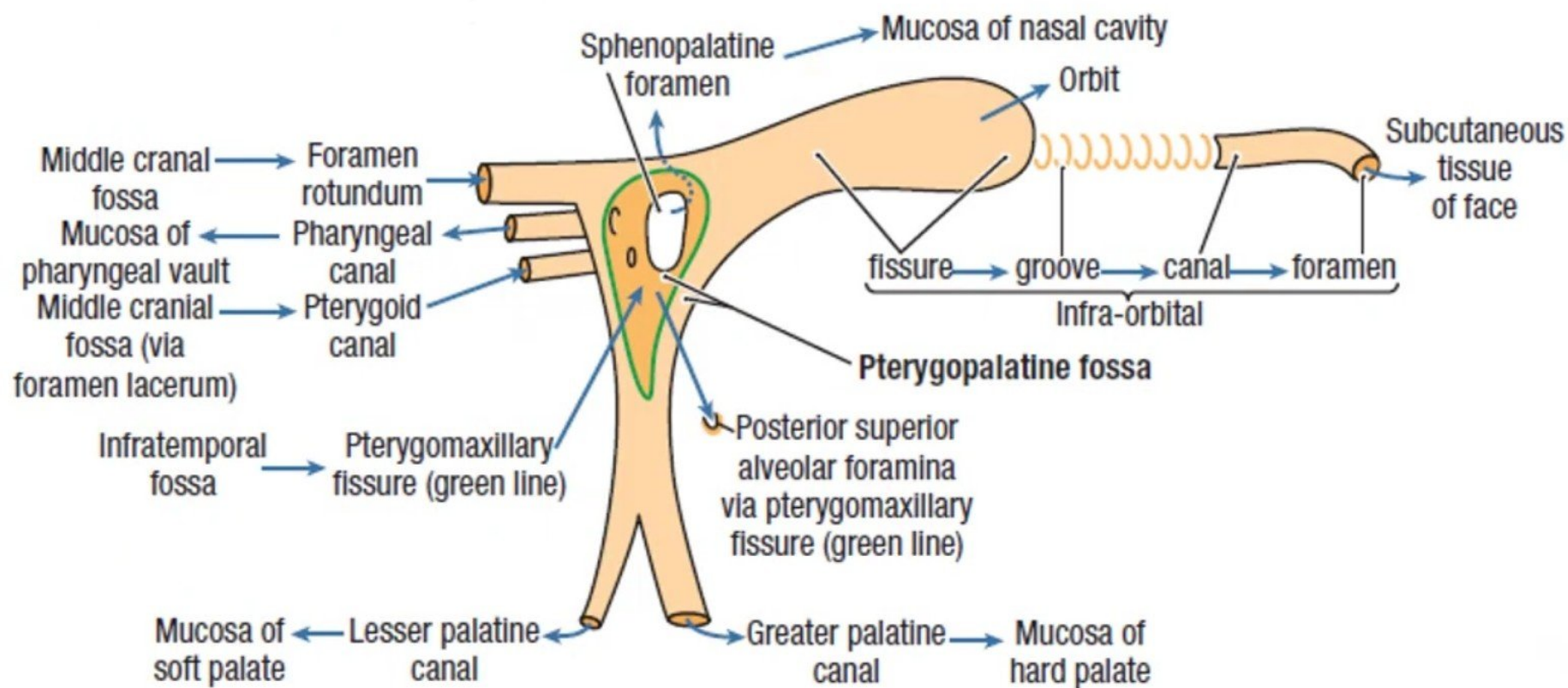
- ▶ From branches of **the maxillary nerve**, with cell bodies in the trigeminal ganglion.

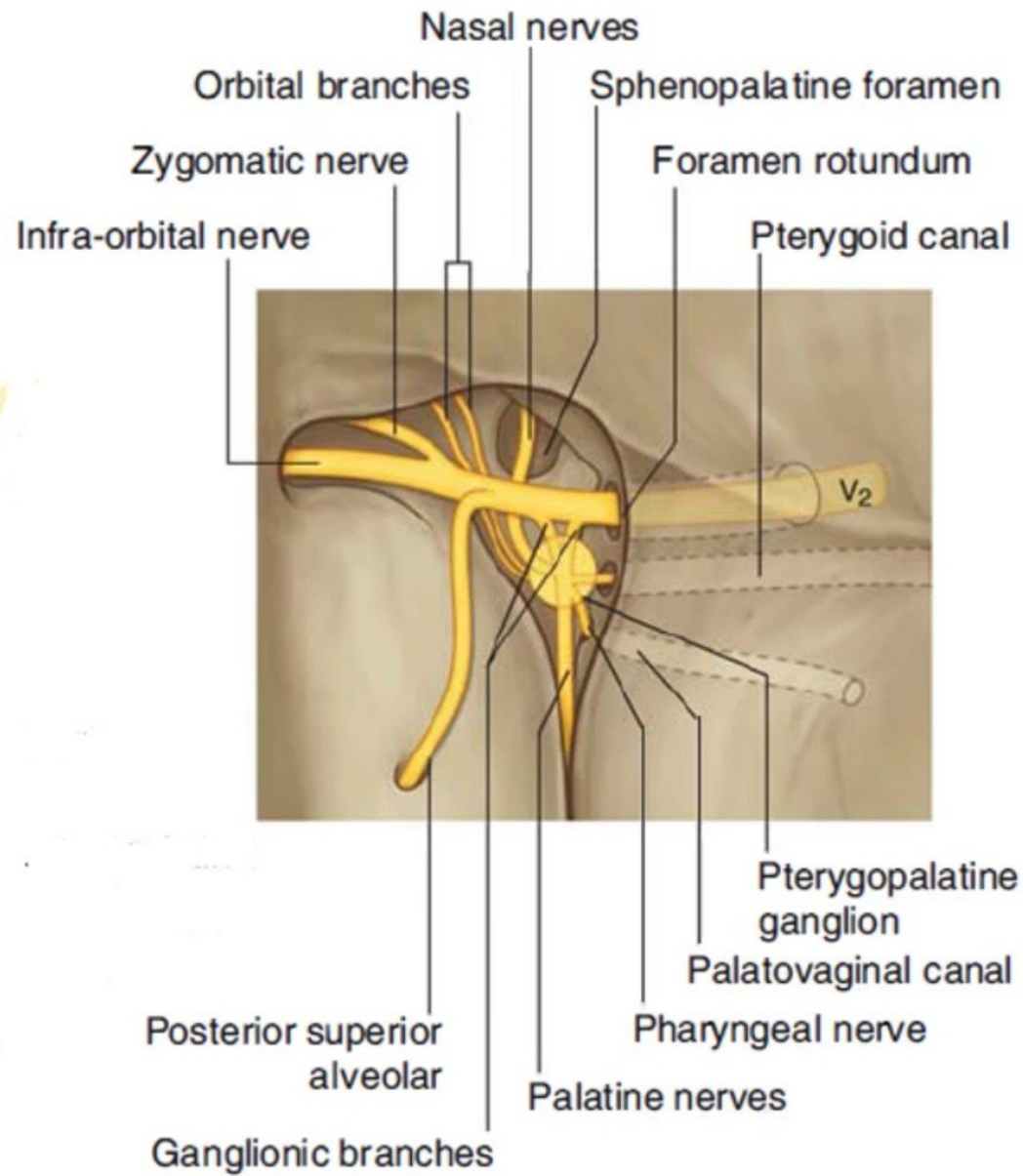
Pterygopalatine ganglion

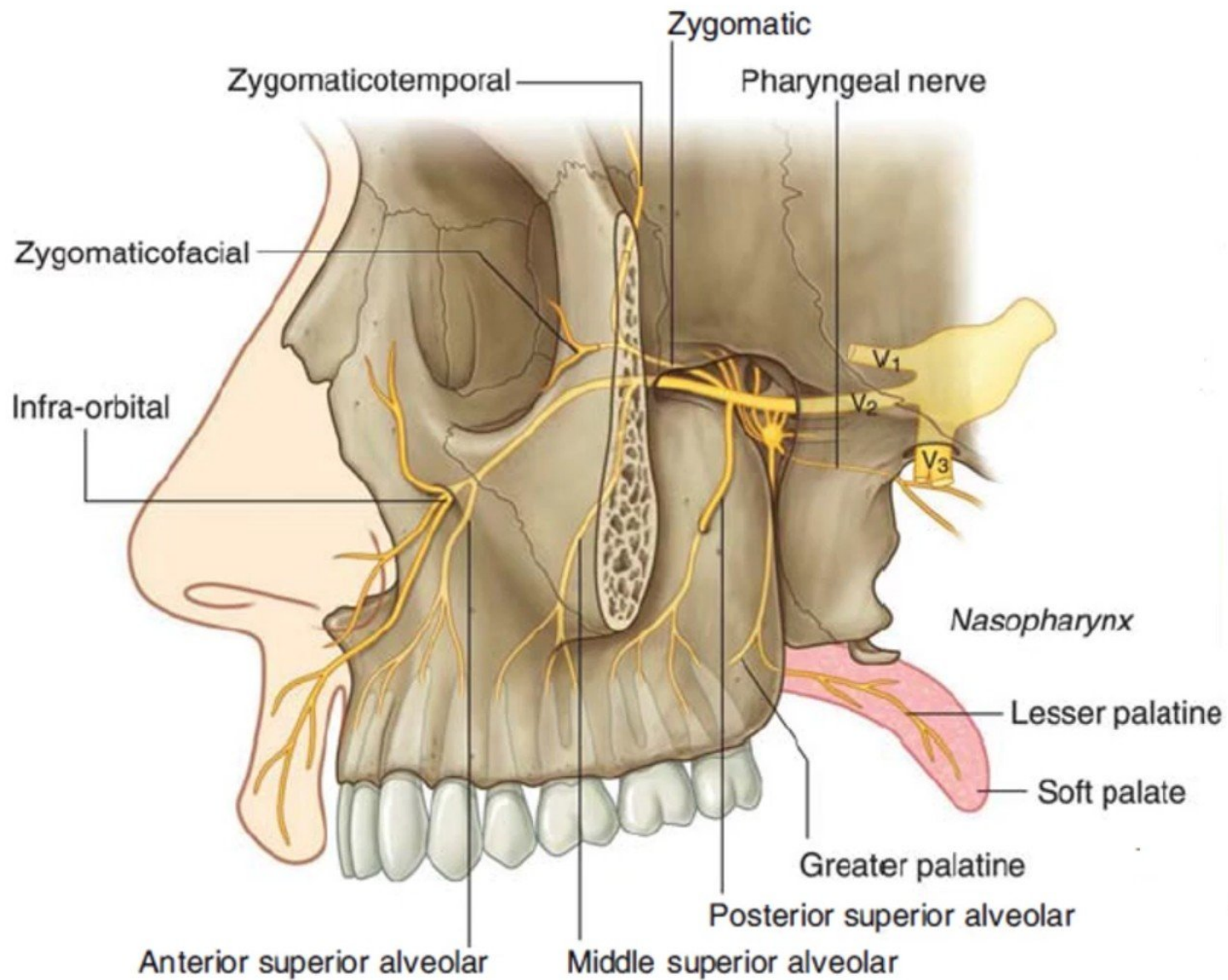
Branches:

- ▶ To the lacrimal gland via the zygomatic and lacrimal nerves, and to mucous glands in the nose, nasopharynx and palate via maxillary nerve branches.
- ▶ A few fibres are taste fibres from the palate, which run in the greater petrosal nerve and have cell bodies in the geniculate ganglion of the facial nerve.









Thanks

